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SESSION I**Session Chair: Dr. Perihan ÇULUN Time: 15:30-16:50****Zoom: <https://us06web.zoom.us/j/86143472834?pwd=PwQqh7bPhzXmjcDyOabbKbHM7Hcjhy.1>****Or****ID: 861 4347 2834****Password: PCFM2026**

Time	Activity	Speaker
15:45-15:55	Thermal Inertia and Indoor Comfort Performance of Bio-Based Building Envelopes under Mediterranean Climate Conditions	Hossein BADI
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16:05-16:15	Control of damage mechanisms under tension shear loading in friction stir spot welding of Aluminium alloy and Steel	Mekri Hichem
16:15-16:25	Theoretical Study of the Adsorption of an Organic Dye on a Metal Surface	Boulanouar MESSAOUDI
16:25-16:35	Density functional theory investigation of Transition Metal Oxides for MEMS/NEMS Applications	Mehmet Gokhan Sensoy
16:35-16:45	Evaluation of current heat storage applications of phase change materials	Perihan ÇULUN
16:45-16:55	Evaluation of thermoelectric module production in light of current literature	Perihan ÇULUN

SESSION II**Session Chair: Dr. Özge Yetiş Time: 15:45-17:05****Zoom: <https://us06web.zoom.us/j/86143472834?pwd=PwQqh7bPhzXmjcDyOabbKbHM7Hcjhy.1>****Or****ID: 861 4347 2834****Password: PCFM2026**

Time	Activity	Speaker
15:45-15:55	X-ray attenuation coefficients of tissue equivalent substances of medical importance	Elif KARAHAN
15:55-16:05	Synthesis Of Novel Biochar Based Magnetic Clay Nanocomposites for New Generation Multifunctional Cotton-Polyester Fabrics	Halil Murat TORAMAN
16:05-16:15	Enhanced Photocurrent Response of ERGO/ZnO@TiO ₂ -PbS/CdSe Quantum Dot-Sensitized Heterostructure under Simulated Solar Light	Mesut ERYİĞİT
16:15-16:25	Minimize entanglement by adding the Dirac delta function to the harmonic oscillator	Ahlem Abidi
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PARTICIPANTS

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FT-005 Evaluation of current heat storage applications of phase change materials

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Abstract— Phase change materials (PCMs) stand out as a significant energy efficiency improvement solution in the field of thermal energy storage due to their high heat storage properties at nearly constant temperatures. These properties allow for more efficient, smaller, and simpler-designed systems compared to sensible heat storage methods. PCMs are being studied for their applications in HVAC integration to reduce building heat loads, in solar energy, in thermal management of electronic systems, and in the textile, automotive, and biomedical sectors. PCM materials integrated into building envelopes and air conditioning systems can significantly reduce annual building energy consumption and improve thermal comfort. Generally divided into three groups – organic, inorganic, and eutectic (paraffins and fatty acids) – PCMs have limitations such as low thermal conductivity, phase separation, supercooling, volume expansion, liquid phase leakage, long-term thermal and chemical stability/instability, and material quantity, necessitating further research. Literature research is therefore focusing on various improvement methods and material modifications to enhance performance. This study reviews the literature on improving the energy storage performance of PCMs, and comprehensively evaluates the methods used, the advantages provided, and the limitations encountered. The aim of this study is to present the current state of PCM-based heat storage applications and to shed light on future research.

Keywords: Latent Heat Storage, LHS, phase change materials, PCM

INTRODUCTION

One of the important solutions to the growing energy crisis and environmental problems is energy storage using PCM materials. PCM energy storage is extremely important for increasing energy storage efficiency. PCM materials have significantly high latent heat storage properties relative to their small volume. The operating temperature at which PCM is used should be, on average, close to the melting point of PCM[1-3]. Thermal energy storage is divided into three classes: sensible, latent, and thermochemical. Sensible heat storage utilizes the specific heat capacity of the material, making it possible to harness the latent heat of the material at almost the same temperature change. However, the 15-20% volume change that occurs in the material during solidification must be taken into account[4]. Energy storage that harnesses latent energy is preferred over other methods because it is more practical and less costly[5]. It is estimated that the latent energy of PCMs is, on average, 2.5-3 times that of water.

PCMs can be used to increase the efficiency of heat storage refrigerators, all solar-powered systems[6], geothermal energy, and all waste heat energy applications. And they are already used in many of these applications. They can be useful in recycling waste heat in industrial systems and in ensuring optimum process temperatures[7]. For optimum performance in systems using PCMs, relatively high melting point, stable melt state, chemical stability, and non-reactivity are important. One of the most significant disadvantages of PCMs in thermal storage processes is their low thermal conductivity coefficient, which ranges from 0.2 to 0.7[4]. To overcome this disadvantage, measures such as the use of metal fins and the addition of nanotechnology are being implemented.

Thermal energy storage using PCMs is an important research topic because it can make a significant contribution to addressing increasing energy dependence. PCMs serve as efficient heat storage devices thanks to their ability to store high latent heats at nearly constant temperatures[8, 9]. This method offers the advantage of storing a higher concentration of heat at a nearly constant temperature compared to sensible heat storage methods, all within a smaller system. Since isothermal heat is stored during the phase transition, system design is also simplified. These methods are prominent in a wide range of applications, including PCs, buildings, heat load reduction in solar energy systems, thermal management of imbalance-correcting electronic batteries, and automotive, textile, and biomedical applications[9, 10]. The use of PCMs in buildings, either integrated with HVAC systems or building envelopes, has reduced annual energy consumption by 15-20%[11, 12]. Their use in improving thermal comfort by correcting thermal imbalances is also being investigated[13]. In the field of cooling, thermal energy storage, low-temperature solid-liquid-ice storage, and applications to reduce peak loads and increase efficiency in air conditioning are being investigated [14, 15].

PCM materials are divided into three classes: organic paraffins and fatty acids, and eutectics obtained from inorganic salts and organic/inorganic mixtures[4]. Each type of PCM undoubtedly has its advantages and disadvantages. It is stated that salt hydrates and organic materials are beneficial below 100 °C, while eutectic materials are advantageous between 100-250 °C[16].

The major limitations in the use of PCM materials include low conductivity, leakage in the liquid phase, supercooling, phase separation, volume change (expansion upon solidification), and thermal/chemical stability constraints[17, 18]. These limitations and gaps necessitate further PCM research. This study examines the thermal storage performance of PCMs. The methods, limitations, and advantages used in these studies are discussed. A review of the current literature is conducted. This study aims to evaluate heat storage applications using PCMs from a holistic perspective.

Although significant research exists on PCM materials, the fact that PCMs come in numerous types—inorganic (salts and salt hydrates), organic compounds like paraffin and fatty acids, and polymeric (PEG)—makes this research need to be further expanded. This article evaluates the thermal storage properties of PCM materials and presents examples of ongoing studies in this field. This article can contribute to future research in the field of PCMs for thermal energy storage and serve as an important reference in various applications.

SOME EXPERIMENTAL STUDIES ON HEAT STORAGE OF PCM

Wire mesh application was performed to enable PCM materials to store or discharge heat faster. In the study, where different numbers of wire mesh layers were used, the melting conditions of PCM without wire mesh were observed. Each experimental environment was tested in at least three different heat flux environments. In time-dependent experiments, the melting process of PCM was analyzed, and the relevant plate temperatures were observed. In experiments using wire mesh, an improvement of 2%-24% was observed with an 8-layer wire mesh, while a 16% improvement was reported for a 4-layer wire mesh. Therefore, it is understood that the use of wire mesh in PCMs significantly affects the melting or solidification processes of the material. To ensure the same conditions were met in the experiments, paraffin wax was melted and solidified, and then allowed to melt during the experiment[19].

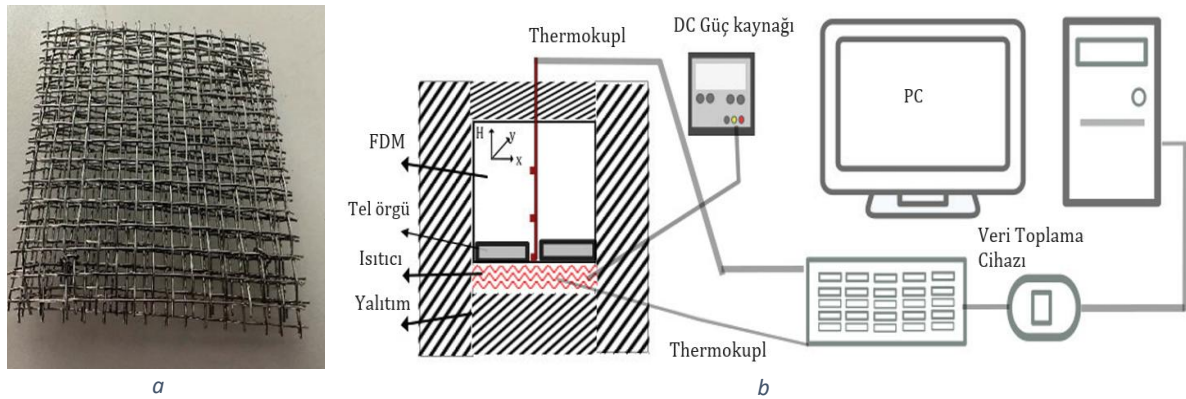


Figure 1. Investigation of the melting and solidification process in PCM material using wire mesh[19]

PCM experimental studies are ongoing, using them in heat exchangers to investigate their efficiency in different heat exchanger structures. These studies investigate parameters such as the melting/freezing processes of PCMs, the effects of different PCM temperatures on efficiency, and the effects of fin number.

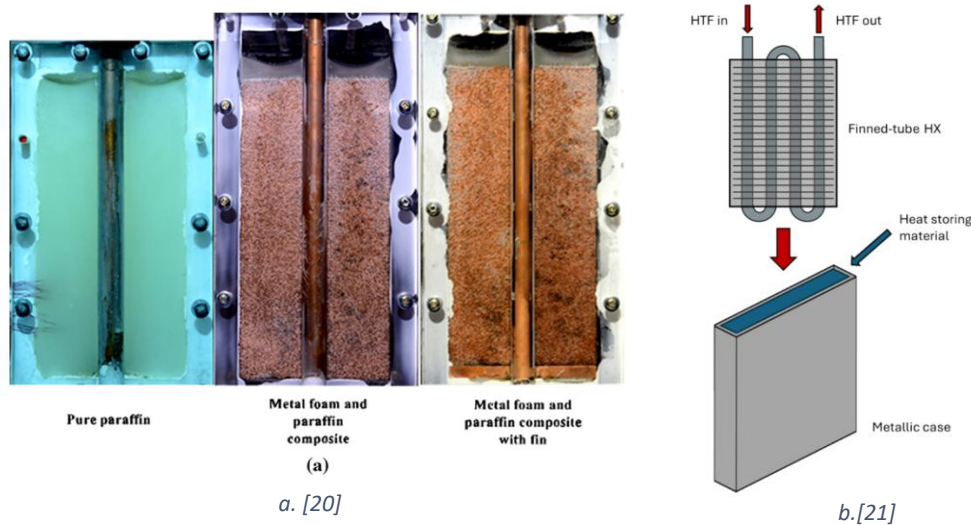


Figure 2. Examples of using PCM in improving heat transfer in heat exchangers

One of the most significant disadvantages of PCMs in thermal storage processes is their low heat transfer coefficient, which ranges from 0.2 to 0.7[4]. To overcome this disadvantage, measures such as the addition of metal fins and nanotechnology are being implemented. Increasing the heat transfer coefficient of PCMs is the focus of many studies. One of these studies involved preparing composite PCMs using paraffin wax and metal foam. The effect of different metal foam ratios on the melting process of the PCM material was investigated, and an increase in heat transfer was observed. In one of these studies, increasing the copper metal foam ratio from 0% to 2.13% reduced the melting time of copper metal foam composite phase change materials from 901 s to 791 s, and increased the heat storage ratio and integrated heat transfer coefficient from 21.81 J/s to 25.08 J/s and from 1.26 W/(mK) to 4.16 W/(mK), respectively. It was stated that while natural convection is the main heat transfer mechanism in the melting process of composite phase change materials with a low copper metal foam ratio, heat conduction plays the primary role in the melting process of composite phase change materials with a high copper metal foam ratio[22].



Figure 1. Increasing the transmission coefficient in PCM materials by using metal foam[22]

In another study examining the use of PCM material with metal foam, it was found that while FDM/wire composite was advantageous in solid and solid-liquid phases, the advantage of metal foam could turn into a disadvantage after all of the FDM melted. The use of foam also reduced the PCM usage ratio by 15%, resulting in a more economical installation setup[23].

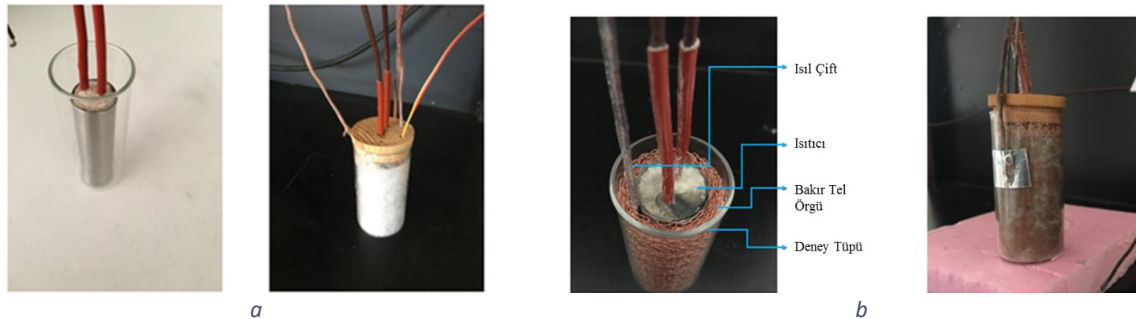


Figure 4. PCM installation setup with metal foam

This study investigated the effectiveness of PCM in the thermal management of portable electronic devices, aiming to regulate the chip temperature of personal digital assistants and computers using a storage unit containing PCM. The results showed that excess heat was effectively absorbed during the study, and that chip temperatures could

be kept below 50°C. Heat source placement, PCM volume, and temperature distribution were found to be important factors[24]. While nanomaterials are used in thermoelectric materials to reduce the thermal conductivity coefficient through doping, PCMs aim to increase thermal conductivity. Experiments are conducted with different ratios of nanopowders or fluids to obtain PCMs with suitable properties, primarily to increase their thermal conductivity coefficient. A key issue in these studies is the precipitation problem. Nanomaterials with higher densities than PCMs have difficulty remaining suspended in the fluid for extended periods[25]. In these types of studies, high concentrations of nanomaterials such as 1%, 2%, 3%, 4%, and 5% are being tested.

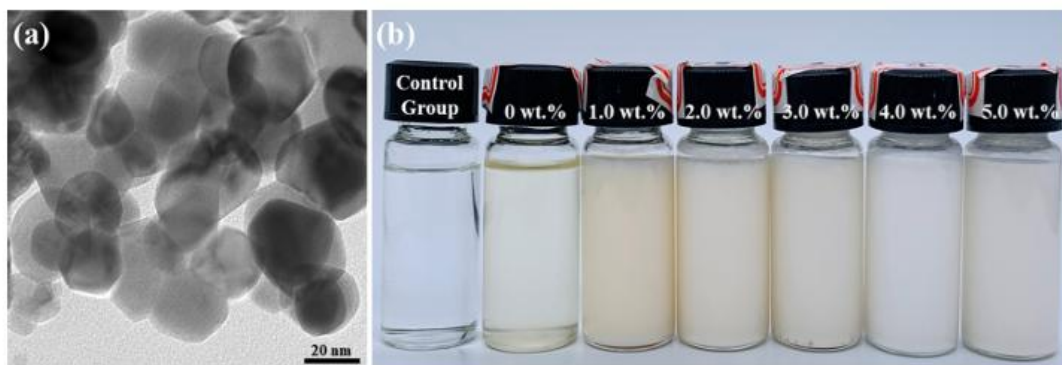


Figure 5. PCM with nano material in %1-5 ratio [25]

In another study on enhancing the thermal conductivity of PCMs, a PCM obtained using alumina (Al_2O_3), a nanomaterial with high thermal conductivity, and MWCNTs, known as multi-walled nanotubes, was investigated. In the study, where Al_2O_3 and MWCNT materials were used at 2%, 4%, and 6% concentrations, the maximum peak temperature obtained after 90 minutes for pure paraffin wax was reported as 61.53°C. For the 2%, 4%, and 6% Al_2O_3 mixture with paraffin wax, the peak temperatures were reported as 62.5, 63, and 64°C, respectively. For the MWCNT mixture, the peak temperatures were 68, 69.86, and 70.55°C, respectively, at 2%, 4%, and 6% concentrations. Increasing the nanoparticle ratio increased the maximum peak temperatures, indicating that the charge and discharge conditions of nano PCMs were further improved. In this case, it was stated that the MWCNT material performed better than the Al_2O_3 material[26].

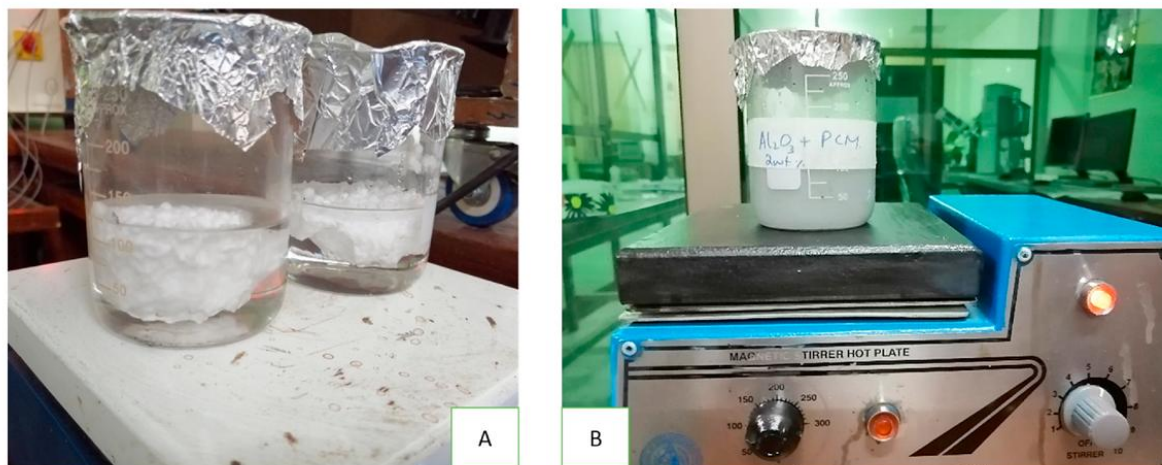


Figure 6. The peak temperatures of Al_2O_3 and MWCNT materials were tested using 2%, 4%, and 6% concentrations [26]

RESULTS AND RECOMMENDATIONS

PCM materials, with their high latent heat storage properties in liquid-mold transitions, are preferred in industry and other places where heat loss occurs, enabling the storage and later use of this heat. Further experimental and numerical studies are crucial for improving the eutectic and nanostructures of these materials, as well as for a thorough analysis of their heat charge-discharge processes. Although these materials are being tested in many places, research into their energy storage properties is extremely important, both due to the high diversity of

materials and the need to improve their characteristics. Heat storage research should examine melting and freezing conditions at different heat fluxes, temperatures, and durations. Indeed, it has been observed that the experiments conducted in this area are insufficient. More numerical studies will undoubtedly be insufficient to fully illuminate this field, as numerical results do not always fully reflect experimental results. Furthermore, numerical results always require time for verification through equivalent experimental studies. In particular, long-term testing of newly developed nano-modified PCMs is of great importance in terms of continuity.

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